
HPL^{ink} Converter

User manual

A user guide to the IPTV channel list and Enigma2 bouquet conversion and management tool.

HPL Converter

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Release date

05.07.2026

Platform

Windows 10 / 11 (desktop application)

Labelling rule: most buttons in the program carry a fixed number in parentheses, e.g. *Load m3u (1)*. This manual refers to those numbers as **(1)**, so a description always points to a specific interface element. The full 1-22 number table is in chapter 14.

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1. About the program

HPL Converter v3.13 [05.07.2026] is a Windows tool that combines the most important TV channel list operations in a single window:

- loading **M3U / M3U8** playlists from a file or directly from a URL,
- logging in to **Xstream Codes** servers and loading Live, VOD and series content,
- **converting a playlist to an Enigma2 bouquet** (type 4097 stream services for satellite receivers),
- sorting channels according to a saved **HPL channel pattern** with intelligent name matching,
- support for **Jason** lists (numbered channel lists in JSON format),
- full list editing: folders, moving, renaming, duplicates, bulk rename, undo,
- a quick **Stream Tester** with colour-coded results,
- a **local M3U TV server** that shares the loaded playlist with a TV on the home network,
- downloading complete **.m3u lists (LIVE + VOD + Series)** via `player_api.php` with series category selection.

The interface is available in **Polish and English** (PL / EN switch in the top right corner) and uses a dark, futuristic "glass cockpit" theme with red tabs and neon cyan accents.

Requirements

- Windows 10 or 11.
- Internet access for the network features (URL, Xstream, list download, stream testing).
- A free port `8899` and firewall permission when the M3U TV server is used.
- A TV / IPTV app on the same local network as the PC when the playlist is shared via M3U TV.

2. Startup and access protection

First launch

1. Run the program (the `hp1_converter.pyw` file or the delivered executable).
2. The main window opens at 1280x880 (minimum 980x620) with the **M3U / URL** tab active.
3. The program starts in Polish. The **EN** button in the top right corner instantly switches the whole interface to English (and back with **PL**).

Note: the program does not persist settings between sessions - after a restart the language, input fields and loaded list return to their defaults.

MAC / MachineGuid protection

The program can be bound to specific computers. Access is controlled by two lists configured in the program distribution (each with up to 5 entries):

- **MAC address list** - at startup the program reads the MAC addresses of the PC's network adapters,
- **MachineGuid list** - the unique Windows identifier from the registry (`HKLM\SOFTWARE\Microsoft\Cryptography → MachineGuid`).

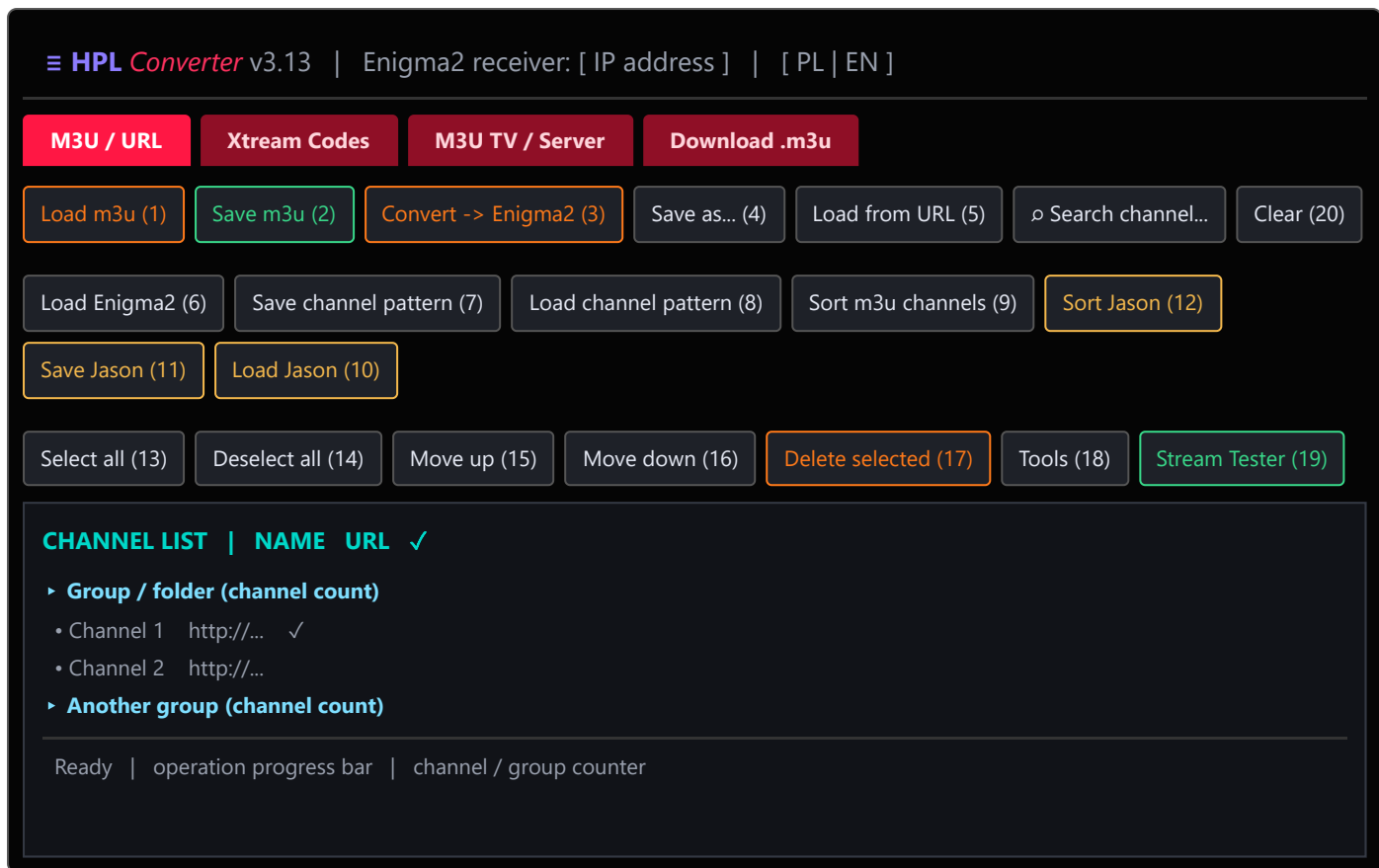
List configuration	Startup condition
Both lists empty	Protection disabled - the program starts on any computer.
MAC list only	At least one of the PC's MAC addresses must be on the list.
MachineGuid list only	The PC's MachineGuid must be on the list.
Both lists filled	Both components must match: MAC and MachineGuid.

If the computer is not authorised, an **"Access denied"** window appears instead of the program, listing the detected MAC addresses and the MachineGuid - send this data to the program vendor to have the computer added to the list.

Reading the MachineGuid: on the client PC run in the command prompt: `reg query HKLM\SOFTWARE\Microsoft\Cryptography /v MachineGuid`

3. Main window

The following drawing is a privacy-safe UI diagram. It contains no IP address, login or user-specific data.



Permanent window elements

- **Program title bar** - logo, full program name and version.
- **IP address (Enigma2 receiver)** - an informational field for the receiver's LAN address (default 192.168.1.4). It serves as a handy note while working with bouquets; conversion and file saving do not send anything to this address.
- **PL / EN** - instant language switch for the entire interface.
- **Four red tabs** - M3U / URL (main list work), Xtream Codes, M3U TV / Server, Download .m3u.
- **Status bar** at the bottom - operation messages on the left, channel and group counter on the right.
- **Slide-in progress panel** - during longer operations (loading, converting, testing, downloading) a panel with the operation name and percentage appears above the status bar.

Channel tree

The list is presented as a tree: **groups (folders)** with a channel counter, and channels with the **NAME, URL** and check-mark ✓ columns. Groups are highlighted in colour and bold. Clicking the ✓ column or double-clicking an entry toggles its check mark; checking a group checks all of its channels.

4. Loading channel lists

Load m3u (1)

1. Click **Load m3u (1)** and open a `.m3u`, `.m3u8` or `.txt` file.

2. The program reads `#EXTINF` entries (name from `tvb-name` or the title, group from `group-title`) and `http/https/rtmp/rtsp` URLs.
3. Channels without a group are placed in the `Ungrouped` folder.

Load from URL (5)

1. Click **Load from URL (5)**. A window with a field for the m3u link opens.
2. The address must start with `http://` or `https://`.
3. The **User-Agent (optional)** field is pre-filled with the identifier of a popular IPTV app - some providers require a specific User-Agent and return an empty response without it.
4. Click **Download**. The download runs in the background (gzip/deflate compression is supported) and the list fills the tree when finished.

Load Enigma2 (6)

Load Enigma2 (6) opens an Enigma2 bouquet / list file (`.tv`, `.radio`, `.txt`, `.ep1`, `.lst`). The program reads service names from the `#SERVICE` / `#DESCRIPTION` lines and places them in the **Enigma2** folder (without URLs). A list loaded this way primarily serves as the **source of a channel-order pattern** - see chapter 6.

Tip: a bouquet file (e.g. `userbouquet.xxx.tv`) can be downloaded from the receiver via FTP from the `/etc/enigma2` directory.

Other sources

A list can also be loaded via the **Xtream Codes** tab (chapter 11) or downloaded in full to a file on the **Download .m3u** tab (chapter 13) and then opened with button (1). Jason lists are described in chapter 7.

Important: loading a new list replaces the current tree content. Unsaved changes are lost.

5. Converting to the Enigma2 format

1. Load

Load an M3U list with (1), (5) or from Xtream (21).

2. Organise

Optionally sort and edit the list (chapters 6-9).

3. Convert

Click Convert -> Enigma2 (3). The progress bar shows the status.

4. Save

Click Save as... (4) and store the bouquet file.

The conversion builds an IPTV bouquet in the Enigma2 format in memory. Every channel becomes a type `4097` stream service:

```
#NAME IPTV
```

```
#SERVICE 4097:0:1:2:0:0:0:0:0:http%3a//stream/address:Channel name
```

#DESCRIPTION Channel name

- Colons in the URL are replaced with `%3a`, as required by the bouquet format.
- Service identifiers are assigned automatically (2, 4, 6, ...).
- **Save as... (4)** stores the conversion result in a text file. Without a prior conversion the button reports "No data to save - convert first".

Using it on the receiver: upload the saved file via FTP to `/etc/enigma2` as e.g. `userbouquet.iptv.tv`, register it in `bouquets.tv` (or use a bouquet editor), then reload the receiver's channel list.

Saving the list as M3U (2)

Save m3u (2) stores the current tree content (after all edits) as a standard `.m3u` file with `#EXTINF` entries and `group-title` groups. This is the recommended way to preserve an organised list for other IPTV apps.

6. HPL channel pattern and sorting

A channel pattern is a stored **order of channel names** that the program can use to sort any other list - even when the names differ in details (HD/FHD/4K, "PL:" prefixes, diacritics and so on).

Creating a pattern

1. Load the reference list with **Load Enigma2 (6)** - e.g. a bouquet with your favourite channel order from the receiver.
2. Click **Save channel pattern (7)**. The program stores the unique channel names in `channel_template.hpl` next to the program and reports the number of saved entries.

Loading a pattern (8)

Load channel pattern (8) opens a previously saved pattern (`.hpl`, `.json` or a plain text file with one channel name per line).

Sorting m3u channels (9)

1. Load an M3U list and a pattern (6 or 8).
2. Select **exactly one group** in the tree (or any channel inside it).
3. Click **Sort m3u channels (9)**.
4. Channels matched against the pattern are placed at the top of the group in pattern order; the remaining ones follow in their previous order.

How name matching works

- Names are normalised: upper case, diacritics removed, tags such as `PL, HD, FHD, UHD, 4K, SD, HEVC, VIP, RAW` and prefixes like `XX:` stripped.

- Name similarity is scored with a text comparison algorithm; the required match threshold is **0.88**, so unrelated channels are not paired.
- **CANAL+** channels have extra rules: numbers (CANAL+ 1, CANAL+ SPORT 3...) and the SPORT / PREMIUM / FILM / SERIALE / DOKUMENT tokens are compared, preventing mix-ups within the channel family.

Tip: a sort operation can be reverted with **Tools (18) — Undo last change**.

7. Jason lists

A Jason list is a numbered channel list in JSON format (`.jason` or `.json` extension) where each entry has the `name`, `link`, `icon` and `target` fields:

```
{ "1": { "name": "Channel", "link": "http://...", "icon": "http://...", "target": "POLSAT" }, "2": { ... } }
```

Operations

- **Load Jason (10)** - opens the file and shows the entries in the **Jason** folder as a numbered list (`001. Name`). Entries are read in numeric order.
- **Save Jason (11)** - stores the current tree order, renumbering the entries 1...N. The icon and target fields are preserved.
- **Sort Jason (12)** - sorts the list according to the loaded HPL pattern (chapter 6). When several entries share the same name, the one with `target = POLSAT` is preferred.

Note: while editing a Jason list, the numbering shown in the tree refreshes automatically after moving, deleting, cutting and pasting entries.

8. Editing the channel list

Selecting

- **✓ column:** clicking the ✓ cell or double-clicking an entry toggles its check mark. Checking a group includes all of its channels.
- **Select all (13) / Deselect all (14)** operate on the whole list.
- Independently of ✓, normal row highlighting works (click, **Ctrl** +click, **Shift** +click) and is used for moving items and the Stream Tester.

Moving and deleting

- **Move up (15) / Move down (16)** - moves the highlighted entries one position within their folder.
- **Delete selected (17)** - deletes highlighted and checked (✓) entries after confirmation. Deleting a group deletes its channels.

Context menu (right mouse button)

Command	Action
Rename	Renames a single channel (dialog with a text field, Enter confirms).
Rename folder	Renames the group containing the selected entry.
Move to folder...	Moves the highlighted channels to a group chosen from a list.
Cut / Paste here	Cuts entries (including whole groups with their content) to the program clipboard and pastes them after the target entry.
Delete	Works like button (17).

Search (20)

The **Search channel...** field in the top right corner of the M3U / URL tab filters the list by a fragment of the channel or group name (confirm with **Enter**). The **Clear (20)** button restores the full list. Edits made on a search result are automatically carried over to the full list.

9. Tools (18)

The **Tools (18)** button opens a menu of housekeeping functions:

Function	Description
Create folder	Adds a new, empty group with the given name.
Merge folders	Moves all channels from a source folder to a target folder (chosen from lists).
Remove empty folders	Removes all groups without channels and reports their number.
Find duplicates	Shows a report of duplicate pairs. A duplicate is detected by normalised name, by URL, or by both.
Remove duplicates	Keeps the first occurrence of each channel, removes further repetitions and reports their number.
Bulk rename	Removes or replaces a given text fragment in all channel names (e.g. stripping a "PL: " prefix). Redundant spaces are cleaned up automatically.
Auto organize folders	Distributes channels into thematic folders based on their names: Sport, Movies, Series, Kids, Documentary, News, Music, Entertainment, Other.
Export report	Saves the list to a <code>.csv</code> file (columns: group, name, url) or the last text report to <code>.txt</code> .
Undo last change	Restores the list state from before the last editing operation (multi-level undo history).

10. Stream Tester (19)

1. Highlight channels or whole groups in the tree.
2. Click **Stream Tester (19)**. The test runs in the background and the progress bar shows the status.
3. Each channel is probed with a short HTTP request (5-second limit) and receives a colour:

Colour	Result	Meaning
green	OK	The server responded correctly within the time limit.
amber	WARN	The server responded with 401 / 403 / 405 (it refuses the probe, but the stream may still play) or responded slowly.
red	FAIL	No response, connection error or empty URL.

When finished, the status bar shows a summary such as `Stream test completed: 42 OK, 3 WARN, 5 FAIL`.

Note: the test only checks the server response - it does not play video and uses negligible bandwidth. A WARN result does not necessarily mean the channel is broken.

11. Xstream Codes (21)

1. Open the **Xstream Codes** tab.
2. Enter the **Host** in the form `http://your-server.com:port`, the **Username** and the **Password** (the password field is masked).
3. Select what should be loaded: **Live TV**, **VOD**, **Series** (at least one).
4. Click **Connect & Load (21)**.

The program logs in via `player_api.php` and, after a successful login, shows a status like `Logged in: name | Expires: date`. Channels are loaded into the main tree (the program switches to the M3U / URL tab automatically) in groups with prefixes:

- `[LIVE] Category name` - live channels,
- `[VOD] Category name` - movies,
- `[SERIES] Category name` - series entries.

A list loaded this way can be edited, sorted, converted to Enigma2 (3), saved as m3u (2) or shared via M3U TV like any other list.

Important: login credentials are not stored anywhere - they must be re-entered after the program is closed. The generated stream URLs contain the account's username and password, so saved m3u files should be handled with care.

12. M3U TV - local server (22)

The **M3U TV / Server** tab shares the loaded channel list on the home network - with a TV, a set-top box or an IPTV app - without copying any files.

1. Load a channel list (M3U / URL tab or Xstream).
 2. Click **Start server (22)**. The program shows the **Server running** status and the playlist address in the form `http://your-PC-IP:8899/iptv.m3u`.
 3. Use **Copy URL** and enter the address in the TV's IPTV app as *"Import m3u-file from URL"*.
 4. The cards below the address show the number of **shared channels** and the **port** (8899).
 5. The same button stops the server (**Stop server (22)**).
- The server always shares the current list state - tree changes (edits, sorting) are visible the next time the playlist is fetched.
 - At the `/status` path the server returns a short diagnostic text with the channel count.
 - Entries without a URL (e.g. loaded from an Enigma2 file) are skipped in the shared list.

Important: the PC and the TV must be on the same local network and the program must keep running. On the first server start Windows Firewall may ask for permission - allow access on the private network.

13. Download .m3u LIVE + VOD + SERIES

The fourth tab downloads a **complete list** from an Xstream server and writes it directly to files on disk: LIVE channels and VOD movies into one file, and series episodes into a separate one. The list is built locally via `player_api.php` - no server-side script is ever executed.

Step by step

1. Login data.

- Fastest way: paste the full playlist address (a `get.php / m3u_plus` link) into the top field and click **Load from URL** - the host, username and password fill in automatically.
- Or enter the **Host** (`http://server:port`), **Username** and **Password** manually.

2. Download scope. Check **Live TV**, **VOD** and/or **Series** (all enabled by default).

3. Series categories. With Series checked, click **Select series categories...**:

- the window fetches the category list from the server,
- type e.g. `PL` in the filter field and use **Select matching** (or **Select all** / **Clear selection**),
- confirm with **Confirm selection**. The label under the button then shows e.g. "Series categories: selected 12 / 96".

4. Output files. Choose the **LIVE + VOD** file (default `playlist.m3u` next to the program) and the **SERIES** file (default `playlist_series.m3u`).

5. Download list. The program verifies the account, then downloads LIVE, VOD and the episodes of the selected series in turn. Progress is shown on the bar and details in the time-stamped log window. The **Cancel** button stops the operation at any moment.

6. Summary. A final dialog shows the number of LIVE / VOD entries and series episodes together with the saved file paths. The finished files can be opened right away with **Load m3u (1)**.

- Groups in the output file: LIVE channels keep their category name, movies get the `VOD -` prefix, series the `SERIES -` prefix.
- Entries include the `tvb-id`, `tvb-name`, `tvb-logo` and `group-title` metadata.
- On connection problems the program automatically retries the requests with alternative User-Agent identifiers.

Note: downloading series requires a separate request for every title, so with many selected categories it may take several minutes. The log reports progress every 200 titles.

14. Button numbering and program files

Button numbering 1-22

No.	Button (EN)	Function
1	Load m3u	Load a list from a .m3u / .m3u8 / .txt file
2	Save m3u	Save the current list to a .m3u file
3	Convert -> Enigma2	Convert the list to an Enigma2 bouquet (4097 services)
4	Save as...	Save the conversion result to a text file
5	Load from URL	Download an m3u list from the internet (with a User-Agent field)
6	Load Enigma2	Load an Enigma2 bouquet / list (pattern source)
7	Save channel pattern	Save the order pattern to channel_template.hpl
8	Load channel pattern	Load a pattern from a .hpl / .json / .txt file
9	Sort m3u channels	Sort the selected group according to the pattern
10	Load Jason	Load a numbered .json / .json list
11	Save Jason	Save the Jason list with fresh numbering
12	Sort Jason	Sort the Jason list according to the pattern
13	Select all	Check ✓ on all entries
14	Deselect all	Remove ✓ from all entries
15	Move up	Move the highlighted entries up
16	Move down	Move the highlighted entries down
17	Delete selected	Delete highlighted and checked entries
18	Tools	Housekeeping menu (chapter 9)
19	Stream Tester	Stream availability test with colour-coded results
20	Clear	Clear the search box and restore the full list
21	Connect & Load	Xtream Codes login and channel loading

No.	Button (EN)	Function
22	Start / Stop server (M3U TV)	Start and stop the local playlist server

Files created by the program

File	Purpose
channel_template.hpl	The channel order pattern saved next to the program with button (7). JSON format: <code>{"format": "HPL_LIST", "source": "ENIGMA2", "channels": [...]}</code> .
playlist.m3u	Default LIVE + VOD output file of the Download .m3u tab (next to the program).
playlist_series.m3u	Default series episodes output file (next to the program).
.m3u, .txt, .csv, .json files	Written only to locations chosen by the user (buttons 2, 4, 11 and Export report).

Note: the program creates no configuration files and no registry entries (apart from reading the MachineGuid for the access protection). Deleting the program folder removes it completely.

15. Troubleshooting

The program does not start - "Access denied" window

The computer is not on the list of authorised machines (chapter 2). Send the detected MAC addresses and the MachineGuid shown in the error window to the program vendor.

Load from URL returns an error or an empty response

1. Check that the address starts with `http://` or `https://` and works in a browser / VLC.
2. Some providers require a specific User-Agent - keep the default value or enter your IPTV app's identifier.
3. Verify that the provider account is active and the concurrent connection limit is not exceeded.

Sorting (9) or (12) does not work

- "Load channel pattern (6 or 8)" - load a pattern first: an Enigma2 bouquet (6) + Save pattern (7), or a ready pattern file (8).
- "Select one M3U group to sort" - select exactly one group in the tree (applies to button 9).
- "No matching channels found" - the list names differ too much from the pattern (match threshold 0.88). **Bulk rename** in Tools helps to unify the names.

"No data to save - convert first"

Save as... (4) stores the conversion result - click **Convert** -> **Enigma2 (3)** first.

The M3U TV server does not start or the TV cannot load the list

1. Load a channel list first - an empty program has nothing to share.
2. Port may be taken by another program - close it or restart the PC.
3. Allow the program through Windows Firewall on the private network.
4. The TV must be on the same network; disable client isolation in the router / guest Wi-Fi.
5. The program must keep running while the list is used on the TV.

The Stream Tester shows only FAIL / WARN

Some IPTV servers reject short probe requests (WARN for 401/403/405 codes) - the channel may still play in a player. Also check the account validity and the provider's connection limit.

Xtream login failed

1. Enter the host with its port, e.g. (no trailing slash).
2. Check the username and password; the message means rejected credentials or an expired account.
3. Select at least one data type: Live TV, VOD or Series.

"Invalid Jason file format"

The file must be valid JSON with the structure described in chapter 7 (a dictionary of numbered entries with the name / link / icon / target fields).